NUSRAT BEGUM

Python Full-Stack Developer | FastAPI, React, TypeScript, PostgreSQL | M.Sc. Computer Science (MUICT)
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SUMMARY

Python full-stack developer with an M.Sc. in Computer Science from Mahidol University (MUICT) and first author of an IEEE-published paper. Builds and deploys business web applications with FastAPI, React and PostgreSQL, including user authentication, validated data entry, automated reporting, CI/CD testing and documented REST APIs.

SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL
- **Backend:** FastAPI, Pydantic, SQLAlchemy, REST APIs, OAuth2, JWT
- **Frontend:** React, Vite, Tailwind CSS, HTML, CSS
- **Databases:** PostgreSQL, SQLite, MongoDB
- **Testing and QA:** pytest, Playwright, Test Planning, User Acceptance Testing (UAT), Accessibility Testing (WCAG)
- **DevOps and Tools:** Git, GitHub Actions (CI/CD), Docker, Docker Compose, Linux, Vercel, pandas, scikit-learn
- **Cloud (currently learning):** Kubernetes, AWS, GCP, Redis, Microservices
- **IT Project Delivery:** Requirements Analysis, Technical Documentation, API Documentation (OpenAPI), Data Reporting

PROJECTS

Smart Calculator – Authenticated REST API with Web Client | FastAPI, SQLAlchemy, PostgreSQL, JWT, Docker | Live demo

- Secured per-user calculation history by implementing the OAuth2 password flow with JWT access tokens and bcrypt password hashing.
- Maintained a releasable codebase with 20 automated pytest tests and a Docker build running in GitHub Actions CI.

Diabetes Risk Prediction API | FastAPI, Pydantic, scikit-learn, GitHub Actions | Live demo

- Deployed a scikit-learn model as a REST API with a web form, scoring ROC AUC 0.83 and 76% accuracy on held-out test data.
- Rejected invalid input before it reached the model by validating every request field with Pydantic schemas, covered by pytest tests in CI.

Bangkok Air Quality Data Pipeline and Dashboard | Python, REST APIs, GitHub Actions, Chart.js | Live dashboard

- Collected 570 days of air-quality data for Bangkok, Beijing and Los Angeles through a scheduled daily GitHub Actions job calling the AQICN API.
- Built reporting views for 7-day trends, monthly averages and air-quality categories, with date-range filters.

RESEARCH EXPERIENCE

Master's Thesis Researcher | Faculty of ICT, Mahidol University (MUICT) | 2024 – 2026 | IEEE paper

- Designed, built and evaluated EADD, a Python drift-detection library for production ML (first author, IEEE JCSSE 2026).
- Designed an adaptive sequential permutation test that confirms a clear drift after 20 of 199 permutations.
- Detected more drift types than the D3 baseline on real-world data streams (sign test, $p = 0.001$).
- Detected all 4 synthetic drift types with zero false alarms, benchmarked against 7 detectors on 13 real-world data streams.

EDUCATION

M.Sc. in Computer Science, Faculty of ICT, Mahidol University (MUICT) | GPA 3.50 / 4.00 | 2024 – 2026
Thesis: Feature Drift Detection via Adversarial Validation

B.B.A. in International Business, Stamford International University, Bangkok | GPA 3.10 / 4.00

PUBLICATION

N. Begum, P. Yamchote, C. Amornbunchornvej and T. Noraset, "Enhanced Adversarial Drift Detection for MLOps Feature Monitoring," Proc. 23rd International Joint Conference on Computer Science and Software Engineering (JCSSE 2026), IEEE. ieeexplore.ieee.org/document/11596765

CERTIFICATIONS AND AWARDS

- Microsoft Certified: Azure Fundamentals (AZ-900)
- Google Data Analytics Professional Certificate
- Hult Prize 2021: Campus Winner and Bangkok Regional Summit Finalist